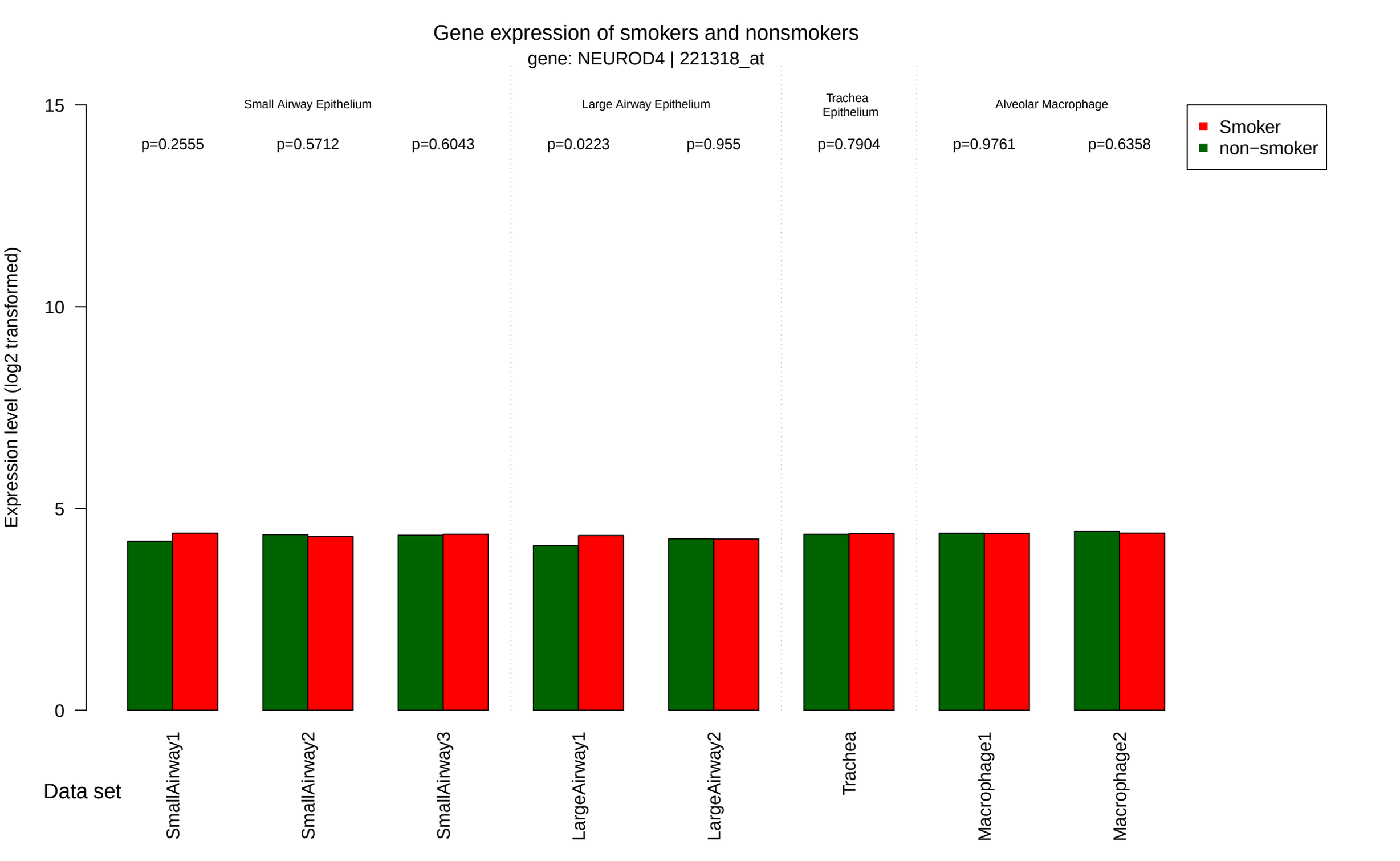
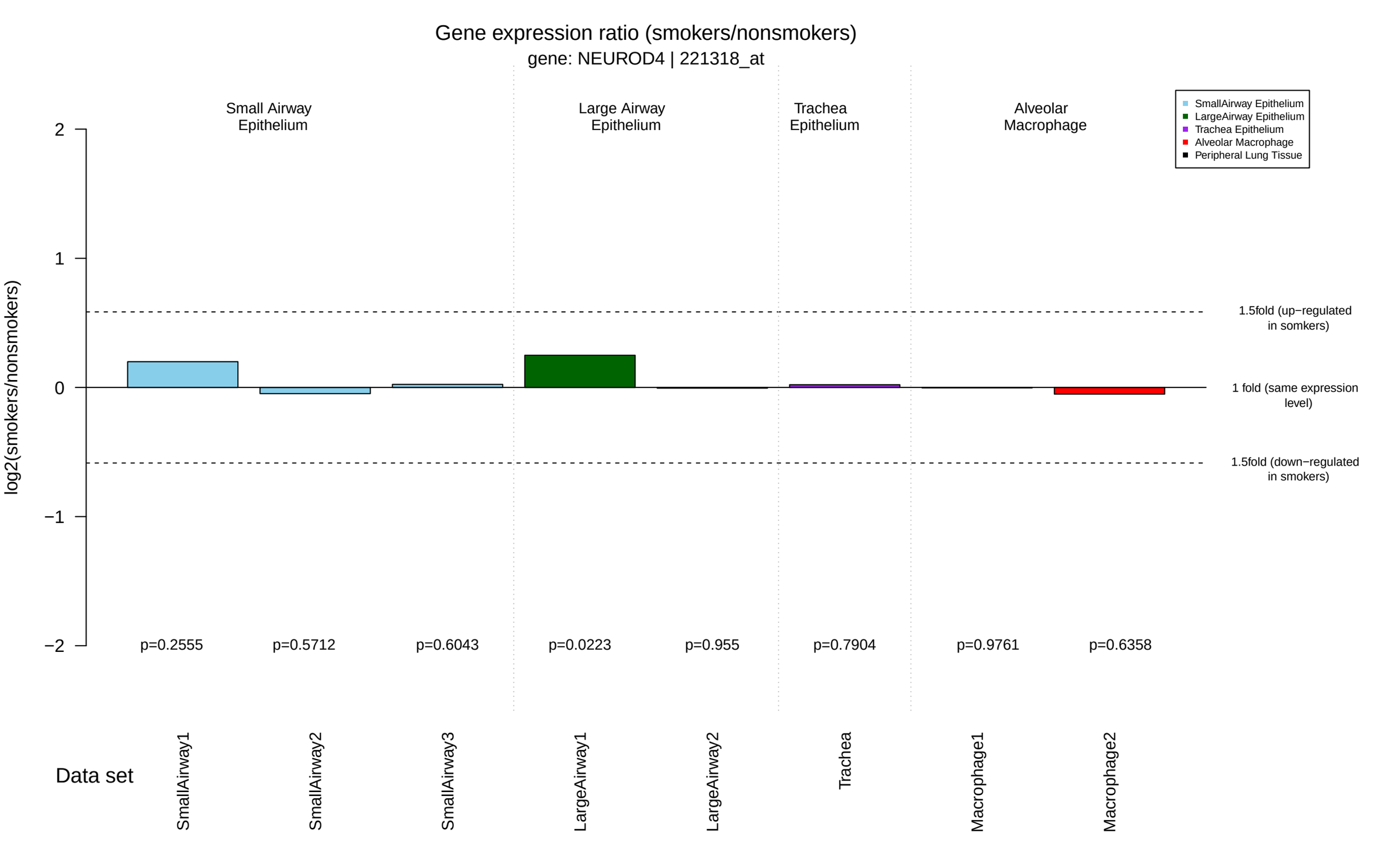
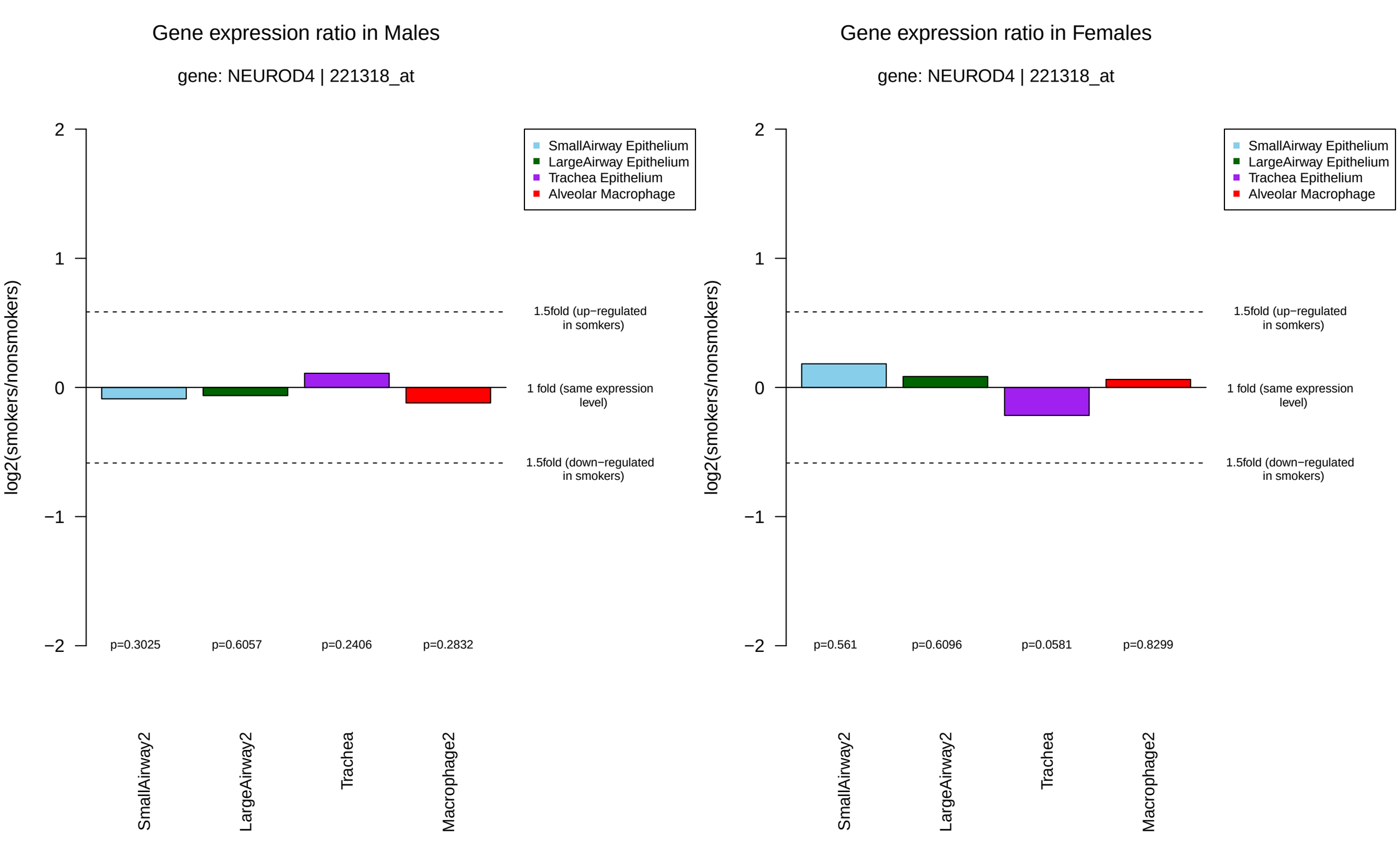
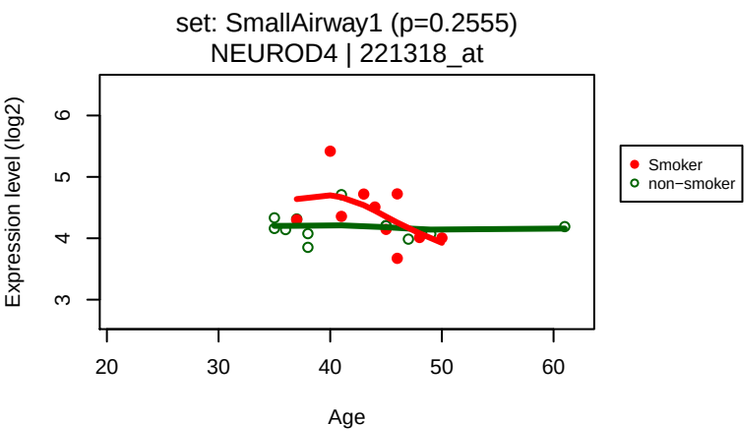
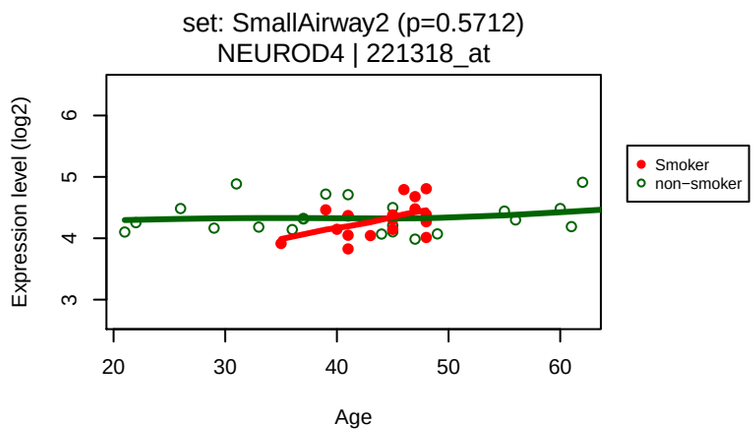
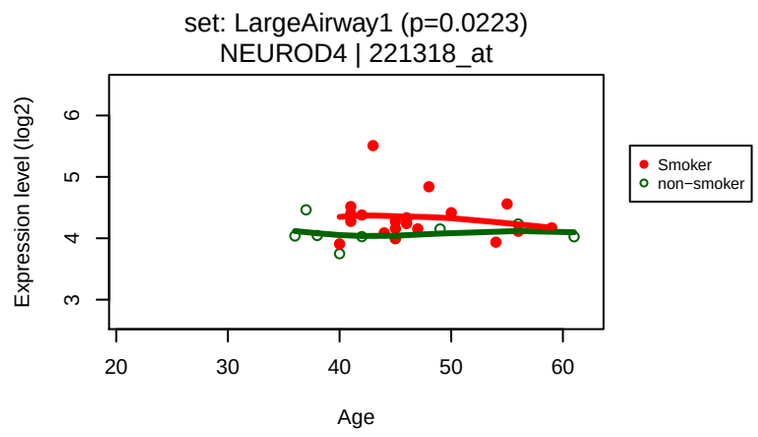

Gene expression of smokers and nonsmokers
gene: NEUROD4 | 221318_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2555
p=0.5712
p=0.6043
p=0.0223
p=0.955
p=0.7904
p=0.9761
p=0.6358
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

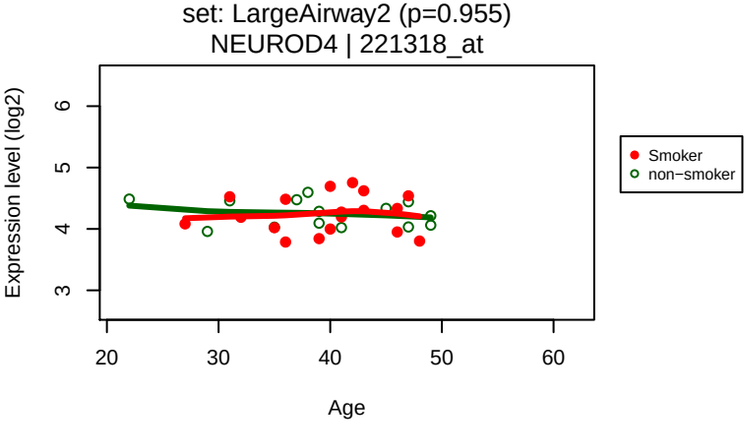

Gene expression ratio (smokers/nonsmokers)
gene: NEUROD4 | 221318_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2555
p=0.5712
p=0.6043
p=0.0223
p=0.955
p=0.7904
p=0.9761
p=0.6358
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

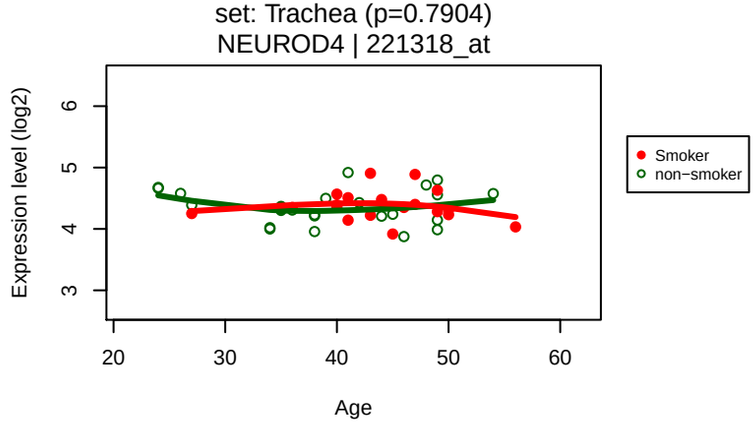

Gene expression ratio in Males
gene: NEUROD4 | 221318_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3025
p=0.6057
p=0.2406
p=0.2832
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NEUROD4 | 221318_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.561
p=0.6096
p=0.0581
p=0.8299
SmallAirway2
LargeAirway2
Trachea
Macrophage2

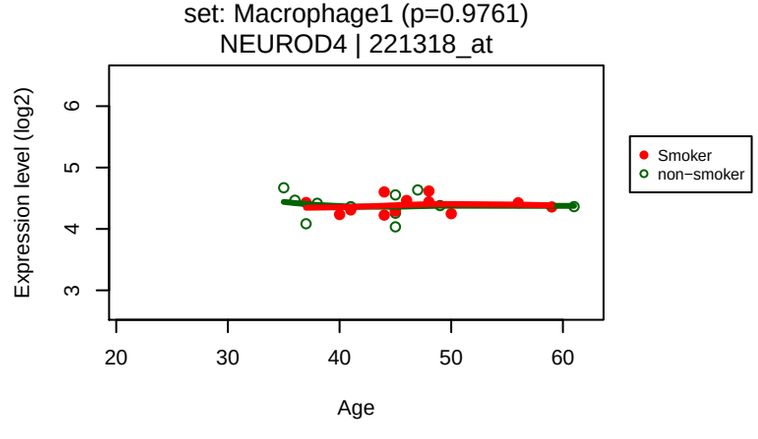

set: SmallAirway1 (p=0.2555)
NEUROD4 | 221318_at
Expression level (log2)
Age
Smoker
non−smoker

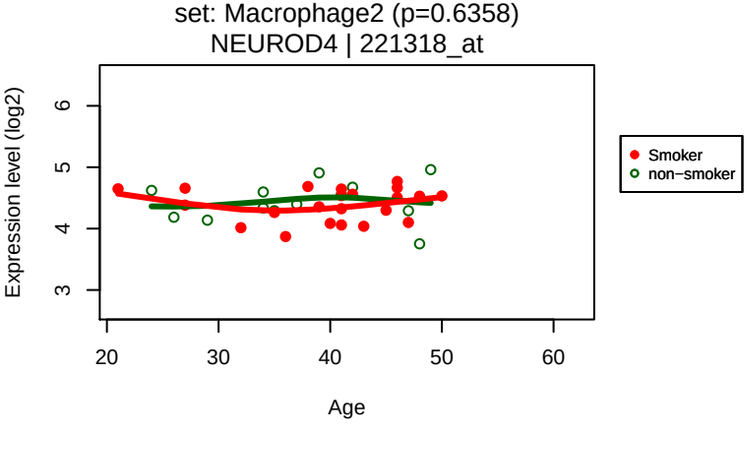

set: SmallAirway2 (p=0.5712)
NEUROD4 | 221318_at
Expression level (log2)
Age
Smoker
non−smoker

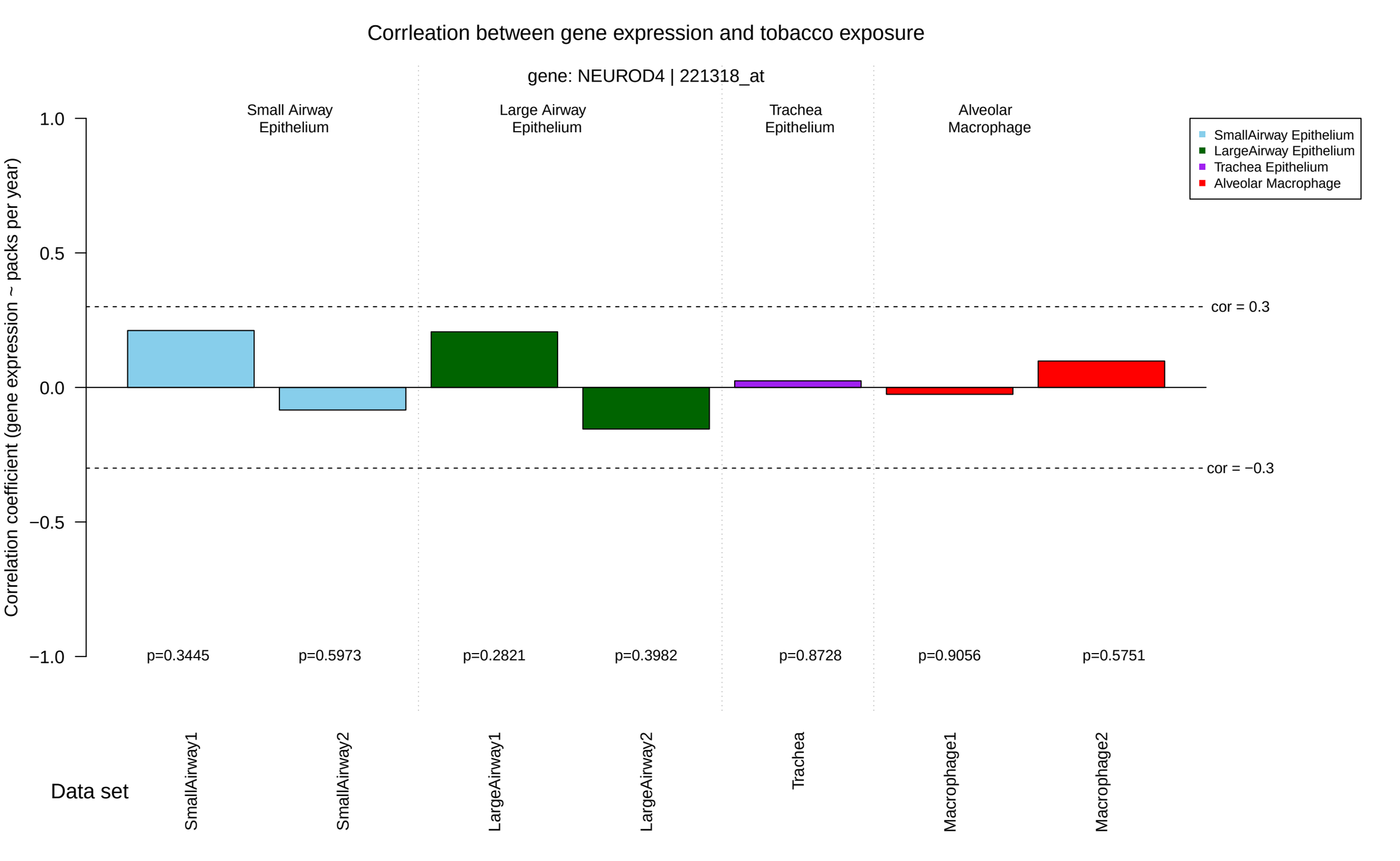

set: LargeAirway1 (p=0.0223)
NEUROD4 | 221318_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.955)
NEUROD4 | 221318_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7904)
NEUROD4 | 221318_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9761)
NEUROD4 | 221318_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6358)
NEUROD4 | 221318_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NEUROD4 | 221318_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3445
p=0.5973
p=0.2821
p=0.3982
p=0.8728
p=0.9056
p=0.5751
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2